WENLI CHEN

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MSc Candidate, Applied Psychology | Human Factors and Human–Computer Interaction

EDUCATION

Shaanxi Normal University Sept 2024 – Jun 2027 (expected)

M.Sc. in Applied Psychology (Engineering Psychology) *Xi'an, China*

Supervisor: Prof. Xiangling Zhuang

Average Grade: 86/100

Awards: Third-Class Graduate Academic Scholarship (2024–2025); First-Class Graduate Academic Scholarship, Top 5% (2025–2026); National Scholarship for Master's Students (selection defence in progress).

Core Courses: Human Factors and Engineering Psychology, Frontiers in Engineering Psychology, User Experience Research Methods, Psychological Statistics and SEM Software Applications, Principles and Techniques of Psychological Measurement

Anhui Normal University Sept 2020 – Jun 2024

B.Sc. in Psychology *Wuhu, China*

Average Grade: 83.4/100

Core Courses: Methodology of Psychology, Experimental Psychology, Experimental Psychology Practicum, Cognitive Psychology, Psychometrics, Advanced Mathematics I and II

RESEARCH INTERESTS

- **Current focus:** Human–AV interaction; understand pedestrians' physiological and psychological processes in AV interactions to improve safety, comfort, trust, and design.
- **General and long-term focus:** Investigate human factors and cognitive processes to improve human performance, experience, and safety across HCI and human–machine systems.

PUBLICATIONS & WORKING PAPERS

- **Chen, W.**, Jiang, Q., Zhuang, X., & Ma, G. (2026). When to brake depends on where to stop: Adaptive timing thresholds for AV yielding behaviour. *Transportation Research Part F: Traffic Psychology and Behaviour, 120*, 103675.
- **Chen, W.**, Jiang, Q., Zhuang, X., & Ma, G. (2025). How deceleration onset time and vehicle speed shape pedestrian perceived safety and comfort in yielding autonomous vehicles. In D. Harris & W. C. Li (Eds.), *Engineering Psychology and Cognitive Ergonomics*. HCII 2025. LNCS, vol. 15776.
- **Chen, W.**, Shen, H., Zhuang, X., & Ma, G. The older, the slower? Adaptive speech-rate design for older adults in personal voice assistant interactions. *International Journal of Human–Computer Interaction*. ***Major Revision***.
- **Chen, W.**, Zhang, L., Ma, G., & Zhuang, X.* "Thank You for Letting Me Pass": Post-Request Feedback Strategies for Autonomous Vehicles Negotiating Right-of-Way with Pedestrians. *International Journal of Human–Computer Studies*. ***Under editorial review***.
- **Chen, W.**[1], Song, Y.[1], Tian, W., Zhuang, X.*, & Ma, G. Direction or Magnitude? eHMI Design for Autonomous Vehicles Based on Kinematic Perception. ***Ready for submission.*** [1]Co-first authors.
- Wu, X., **Chen, W.**, & Zhuang, X.* Effects of Sound Source Location on Attentional Capture by In-Vehicle Voice Warnings. ***Ready for submission.***

CONFERENCE PRESENTATIONS

- **Chen, W.** et al. (2025). How deceleration onset time and vehicle speed shape pedestrian perceived safety and comfort in yielding autonomous vehicles. Oral presentation at HCI International 2025 (HCII 2025), the 27th International Conference on Human–Computer Interaction.
- **Chen, W.** Oral presentation at the Shaanxi Normal University Annual Academic Conference (Second Prize).

SKILLS

- **Human Factors Research:** Experimental protocol design, data collection, surveys, interviews, quantitative and qualitative data analysis, and wearable physiological sensors (ECG, EMG, GSR, and eye tracking).
- **Experimental Platforms and Measurement:** Unity-based traffic and HCI experiments in VR, projection-based, and desktop environments; synchronized collection of behavioral measures, questionnaires, ECG, and EDA.
- **Software & Programming:** SPSS; JASP; PsychoPy; E-Prime; Python (data processing, physiological signal analysis, modeling, and control-system design); RStudio (R for data cleaning, processing, statistical modeling, and visualization); Unity (C# for VR, projection-based, and driving-simulation development); draw.io (scientific diagrams and flowcharts).
- **Languages:** Mandarin Chinese (native); English (proficient; IELTS 7.0: L 8.5, R 7.5, W 6.5, S 6.0).

RESEARCH EXPERIENCE (Project details and demos can be seen here)

Work Area 1: Human Factors and User Experience in Pedestrian–AV Interaction

Research supported by the National Natural Science Foundation of China (NSFC): "Impact of Interaction Strategies Between Autonomous Vehicles and Pedestrians on Pedestrian Crossing Safety and Experience" (PI: Prof. Xiangling Zhuang).

Study 1: Pedestrian-Centered Yielding Behavior of Autonomous Vehicles *2024–2025*

- Addressed the overlooked roles of subjective experience and stopping distance in AV yielding through projection-based and immersive VR experiments.
- Implemented a parametric AV-yielding design varying approach speed, brake-onset TTA, and stopping distance to assess perceived safety, comfort, and deceleration detection.
- Found that later and stronger braking was detected faster but reduced perceived safety and comfort, with these effects amplified when AVs stopped closer to pedestrians.
- Established a 2.5-s brake-onset TTA baseline at a fixed stopping distance and distance-adaptive thresholds for pedestrian-friendly AV yielding.

Status: Published as a journal article in Transportation Research Part F: Traffic Psychology and Behaviour and as an HCII 2025 conference chapter; presented orally at HCII 2025.

Study 2: Post-Request Feedback in AV–Pedestrian Right-of-Way Negotiation *2025–Present*

- Developed gratitude-based post-request feedback strategies to acknowledge pedestrians' cooperation after AV right-of-way requests.
- Evaluated feedback strategies and vehicle appearance anthropomorphism through video-based and immersive VR experiments.
- Found that gratitude-plus-goodwill feedback produced the most favorable subjective experience and future cooperation, while anthropomorphic cues improved subjective evaluations but not immediate yielding behavior.
- Proposed combining indirect requests with gratitude and goodwill feedback for socially responsive AV right-of-way negotiation, with anthropomorphic cues serving as supportive signals for interaction experience.

Status: First-author manuscript under editorial review at International Journal of Human–Computer Studies.

Study 3: Kinematic eHMI Design for Autonomous Vehicles *2026*
Supported by Changan Automobile

- Found that pedestrians were less sensitive to vehicle acceleration than deceleration, identifying a perceptual limitation in interpreting approaching vehicles' motion.
- Proposed three kinematic eHMI designs to address this limitation: speed-change direction cues, speed-magnitude cues, and composite cues combining both types of information.
- Found greater discrimination gains for direction than magnitude cues and better comprehension of composite than magnitude cues, providing evidence to guide future kinematic eHMI design.

Status: Co-first-author manuscript ready for submission.

Study 4: AV–Pedestrian Interaction in Shared Spaces *2025–Present*
Master's thesis research

- Drawing on proxemics, investigated how distance communicates risk in shared-space interactions with passenger AVs and autonomous delivery vehicles.
- Built a Unity VR experiment varying vehicle type, speed, and lateral distance; synchronized BioRadio ECG and GSR recording with subjective safety, comfort, and risk ratings.
- Identified lateral distance as the primary determinant of subjective risk, with speed amplifying risk and vehicle type shaping acceptable proximity.
- Derived vehicle- and speed-specific lateral-distance guidance: 2.0–4.0 m for passenger AVs and 1.5–3.0 m for delivery AVs across 10–30 km/h.

Status: First experiment completed; follow-up study in preparation.

Work Area 2: Age-Friendly Human–Computer Interaction

Adaptive Speech-Rate Design for Older Adults in Voice Assistant Interaction *2025–2026*

- Recruited 67 community-dwelling older adults aged 60–84 to examine heterogeneity in speech-rate needs using a Wizard-of-Oz personal voice assistant paradigm.
- Parametrically varied speech rate (195–285 wpm) and task error sensitivity, assessing comprehension accuracy and interaction satisfaction across later adulthood.
- Found that adults aged 60–75 performed best at 225–255 wpm, whereas those aged 76+ benefited from rates near 200 wpm, particularly in error-sensitive tasks.
- Developed model-based prediction maps linking age, speech rate, and task context to guide adaptive voice-output design.

Status: First-author manuscript under major revision at International Journal of Human–Computer Interaction.